535/2
PHYSICS
Paper 2
July – August 2026
2 Hours

UGANDA MUSLIM TEACHERS' ASSOCIATION

UMTA JOINT MOCK EXAMINATIONS – 2026

Uganda Certificate of Education

PHYSICS

Paper 2

2 Hours

INSTRUCTIONS TO CANDIDATES:

- *This paper consists of **two** examination items.*
- *Answer only **one** item.*
- *Any additional item answered will **not** be scored.*
- *Candidates are **not** allowed to start working with the apparatus for the **first quarter (15minutes) of an hour**. This time is given to enable candidates; read the items thoroughly, checking for the apparatus they will need and plan appropriately.*
- *Mathematical tables and silent non-programmable calculators may be used.*

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Item 1.

Your friend has been shortlisted for a job interview in a company that makes spring pens. The interview assignment is that your friend establishes the stiffness levels of the springs that have been purchased before they are used. The quality standards officer at the company has recommended that the suitable stiffness levels of the required springs range from force constant of **18 Nm^{-1}** to **25 Nm^{-1}**

Task:

You have been provided with one of the springs from the purchased springs among other materials. Help your friend with the interview assignment.

Item 2.

A recent surge in electricity in the area where your relative lives burnt the fuse in the extension cable plug that was being used to protect the electrical devices in the house. Your relative has been advised that an appropriate length (about **6cm**) of a good conducting material of resistance ranging from **0.2 Ω** to **1.2 Ω** may be used to repair the extension cable plug. You have been approached with a sample of the available conducting cables.

Task:

Carry out a scientific investigation to ascertain the appropriateness of the recommended length for use in the plug.

*(Include the following in your report: Aim, variables (independent, dependent and controlled), hypothesis, procedure, errors anticipated, precautions to mitigate the errors, present data accurately (at least **5** sets), analyse data using graph and advice/recommendation).*

END

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